RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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III Semester B.E. Regular / Supplementary Examination February/March 2026

Common to EC/EE/ET

Linear Algebra, Fourier Transforms And Statistics

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator & handbook for mathematics is permitted.

Part A

Question No	Question	M	CO	BT
1.1	The dimension of the vector space spanned by the vectors $(\sqrt{2}, \sqrt{3}, 0), (0, 0, 1), (0, \sqrt{3}, 0)$ IS _____.	02	1	1
1.2	In the vector space R^2, give (i) the rotation matrix through an angle of 60^0 clockwise. (ii) the reflection matrix about the line $y = x$.	02	2	1
1.3	If $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$, then the sum of the eigenvalues and the product of the eigenvalues of A are _____ and _____ respectively.	02	1	1
1.4	Orthogonalize the basis $\{(1, -2), (1, -1)\}$.	02	2	1
1.5	The Fourier coefficient a_0 for the signal $f(t) = 2t - t^2, 0 < t < 3$ is _____.	02	1	2
1.6	The fundamental period of the signal $f(t) = 3\sin\left(\frac{t}{5}\right) + 3\cos\left(\frac{t}{5}\right)$ is _____.	02	1	1
1.7	If $F(e^{-x^2/2}) = \sqrt{2\pi}e^{-s^2/2}$, then $F[e^{-8x^2}]$ =_____ and $F[e^{-(x-3)^2/2}]$ =_____.	02	1	1

1.8	If $F[e^{-2x^2}] = \frac{\sqrt{\pi}}{\sqrt{2}} e^{-\frac{s^2}{8}}$, then $F[e^{-2x^2}\sin 3x]$=_____ and $F[e^{-2x^2}\cos 3x]$=_____.	02	1	1
1.9	For the lines of regression $y = 0.516x + 33.73, x = 0.512y + 32.52$, find the means of x and y.	02	1	1
1.10	The first two moments of a distribution about the value 4 of the variable are -1.5 and 17. Find the mean of the distribution and first moment about the mean.	02	1	1

Part B

Question No	Question	M	CO	BT
2a	Determine whether the vectors $v_1 = (1,2,1), v_2 = (2,3,1), v_3 = (3,5,2)$ are linearly independent. If they are linearly dependent, find the linear dependence relation.	05	2	3
2b	Determine the linear transformation defined by $T : \mathbb{R}^3 \to \mathbb{R}^3$ such that $T(1,1,1) = (1,1,1)$, $T(1,2,3) = (-1,-2,-3)$ and $T(1,1,2) = (2,2,4)$.	05	2	3
2c	For $A = \begin{bmatrix} 2 & -1 & 1 \\ 4 & -2 & 2 \\ -2 & 1 & -1 \end{bmatrix}$, find (a) a basis for the row space of A. (b) the dimension of the left null space.	06	3	3
3a	Decompose the matrix $A = \begin{bmatrix} 5 & 0 & 4 \\ 0 & 5 & 4 \\ 4 & 4 & 9 \end{bmatrix}$ as $A = PDP^{-1}$, where D is a diagonal matrix and P is the diagonalization matrix.	10	3	4
3b	Find the equation $y = mx + c$ of the least-squares line that best fits the data points (1,1), (2,2), (3,2) and (4,4).	06	2	3
	OR			
4a	Apply Gram-Schmidt process to construct an orthonormal basis for the subspace $W = span\,(x_1, x_2, x_3)$ of R^4, where $x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} -2 \\ 2 \\ 3 \\ 5 \end{bmatrix}, x_3 = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}$.	06	2	3
4b	Factorize the matrix $A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ as $A = U\Sigma V^T$ using singular value decomposition process.	10	3	4
5a	An alternating current after passing through a rectifier has the form $I = \begin{cases} I_0 \sin\theta, & 0 < \theta < \pi \\ 0, & \pi < \theta < 2\pi \end{cases}$, where I_0 is the max current. Express I as Fourier series in $(0, 2\pi)$. Also, sketch the graph of the function f(x).	10	2	4

5b	Find the complex form of the Fourier Series of $f(x) = e^{ax}$ over the interval $(-\pi, \pi)$.	06	4	3

OR

6a	The following table gives the variations of periodic current over a period $\begin{array}{c\|c\|c\|c\|c\|c\|c\|c} t & 0 & \frac{T}{6} & \frac{T}{3} & \frac{T}{2} & \frac{2T}{3} & \frac{5T}{6} & T \\ \hline A & 1.98 & 1.30 & 1.05 & 1.30 & -0.88 & -0.25 & 1.98 \end{array}$ Show by harmonic analysis, that there is a direct current part of 0.75 amp. Also find the Fourier series as far as the 2nd harmonics.	10	4	4
6b	Find the half range sine series of $f(x) = \begin{cases} \frac{1}{4} - x, & 0 < x < \frac{1}{2} \\ x - \frac{3}{4}, & \frac{1}{2} < x < 1 \end{cases}$.	06	2	2
7a	Find the Fourier transform of $e^{-a\|x\|}$, where $a > 0$.	08	3	3
7b	Determine the Fourier sine and cosine transform of $e^{-x}\cos 2x$.	08	3	3

OR

8a	Obtain the Fourier transform of the function $f(x) = \begin{cases} a^2 - x^2, & \|x\| \le a \\ 0, & \|x\| > a \end{cases}$. Hence deduce that $\int_0^\infty \left(\frac{\sin t - t\cos t}{t^3}\right) dt = \frac{\pi}{4}$.	08	3	3
8b	Find Fourier sine transform of $f(t) = e^{-t}$. Hence prove that $\int_0^\infty \frac{t^2\, dt}{(t^2+1)^2} = \frac{\pi}{4}$ using Parseval's identity.	08	3	3
9a	The distance (in km) of 60 engineers from their residence to their place of work were found as follows: $\begin{array}{l\|c\|c\|c\|c\|c\|c} \text{Distance (in km)} & 0\text{-}5 & 5\text{-}10 & 10\text{-}15 & 15\text{-}20 & 20\text{-}25 & 25\text{-}30 \\ \hline \text{No. of Engineers} & 8 & 11 & 15 & 12 & 9 & 5 \end{array}$ Compute first four moments about the mean and also find the measures β_1 and β_2 for the given distribution and comment on the nature of distribution.	08	4	4
9b	Ten competitors in a musical test were ranked by the two judges x&y in the following order: $\begin{array}{l\|c\|c\|c\|c\|c\|c\|c\|c\|c\|c} \text{Rank by } x & 1 & 6 & 5 & 10 & 3 & 2 & 4 & 9 & 7 & 8 \\ \hline \text{Rank by } y & 3 & 5 & 8 & 4 & 7 & 10 & 2 & 1 & 6 & 9 \end{array}$ Calculate the co-efficient of rank correaltion.	08	2	3

OR

10a	The table shows the acres x of wheat planted and acres y of wheat harvested, in millions of acres, in a particular country in seven successive years. x: 75.6, 78.8, 78.6, 80.4, 81.8, 78.3, 93.5 y: 68.8, 69.3, 70.9, 73.6, 76.3, 72.6, 86.5 Determine the equations of lines of regressions and the coefficient of correlation for the given data.	08	2	3
10b	The nursing instructor wishes to see whether a student's grade point average (GPA) and age are related to the student's score on the state board nursing examination. She selects five students and obtains the following data. GPA x_1: 3.2, 2.7, 2.5, 3.4, 2.2 Age x_2: 22, 27, 24, 28, 23 State board score y: 550, 570, 525, 670, 490 Fit the best multilinear model that represents the relationship between GPA, age and the student's score.	08	4	4